

Erdős Problem #327

EP-327

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

Suppose $A \subseteq \{1, \dots, N\}$ is such that if $a, b \in A$ and $a \neq b$ then $a + b \mid ab$. Can A be 'substantially more' than the odd numbers?

What if $a, b \in A$ with $a \neq b$ implies $a + b \mid 2ab$? Must $|A| = o(N)$?